Customer Segmentation with Clustering Report

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Problem Statement

The company operates a transnational e-commerce business across five continents and requires customer segmentation to enable more targeted and efficient marketing. The objective is to group customers into meaningful segments based on purchasing behaviour and value characteristics, supporting improved customer lifetime value and early assignment of new customers to appropriate segments.

Methods

Domain Context

Domain guidance suggests a focus on aggregated features (frequency, recency, CLV, average unit cost, and customer age) as the most efficient first step in producing meaningful segmentation.

The Dataset

The dataset contains 951,668 rows across 25 features, representing individual product orders between 1 January 2012 and 30 December 2016. Data cleaning included regex-based type conversion, imputation of missing location values using “unknown”, and removal of 21 duplicate rows.

Feature Engineering

Data were aggregated by customer ID to produce one record per customer. Five engineered features were derived in line with the domain expert’s brief and used as inputs to the clustering models (Table 1).

Feature	Outlier Definition	Description
Frequency	Frequency = sum(Order ID)	Number of orders made by customer over the period of the dataset.
Recency	Recency = End Date – max(Order Date)	Time since customer’s most recent order.
Customer Lifetime Value (CLV)	CLV = sum(Profit)	Total profit generated by customer.
Average Unit Cost (AUC)	AUC = mean(Unit Cost)	Average cost of unit’s purchased by customer.
Customer Age	Age = (End Date) – (Customer Birth Date)	Customer’s age.

Table 1 - Engineered Features

Distribution Analysis

Most engineered features exhibit right-skewed distributions, particularly frequency, CLV, and average unit cost. Recency shows structured tail behaviour, while customer age is approximately symmetric. Outliers were retained, as they may represent high-value or atypical customers relevant to segmentation.

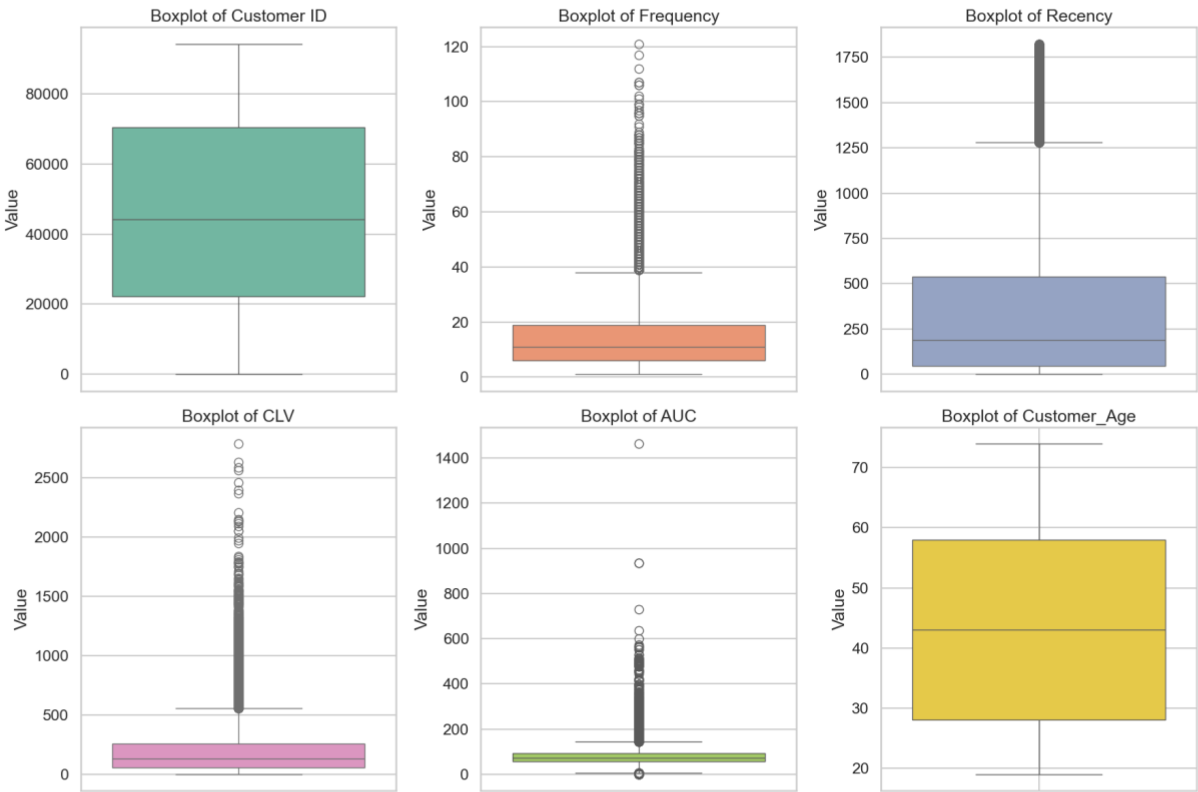

Figure 1 – Boxplot Analysis of Engineered Features

Feature Transformation and Scaling

A Log(1 + x) transformation was applied to right-skewed features to reduce the influence of extreme values on Euclidean distance calculations. All features were standardised using StandardScaler (Table 2).

Feature	Distribution Characteristics	Transformation Applied	Scaling Method	Rationale
Frequency	Strong right skew, long tail	log(1 + x)	StandardScaler	Reduce dominance of high-frequency customers.
Recency	Right skew with structured tail	log(1 + x)	StandardScaler	Stabilise variance and improve distance-based clustering.
CLV	Strong right skew, extreme values	log(1 + x)	StandardScaler	Prevent high-value outliers dominating clusters.
AUC	Right skew, long tail	log(1 + x)	StandardScaler	Compress cost extremes and align scale.
Customer Age	Approximately symmetric	None	StandardScaler	Preserve original distribution while standardising scale.

Table 2 – Transformation and Scaling of the Engineered Features

Optimising Cluster Number k

The optimal number of clusters (k) was determined using converging evidence from the Elbow method, Silhouette score, and Hierarchical clustering applied to the transformed and scaled feature set.

Elbow Method

The elbow plot shows a steep reduction in inertia up to k=4, followed by diminishing returns and a near-linear tail beyond k=8. This suggests a small number of clusters (k=3-5) is optimum.

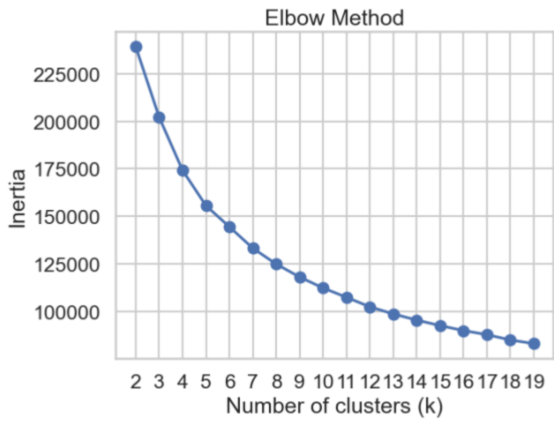

Figure 2 - Elbow Method

Silhouette Score

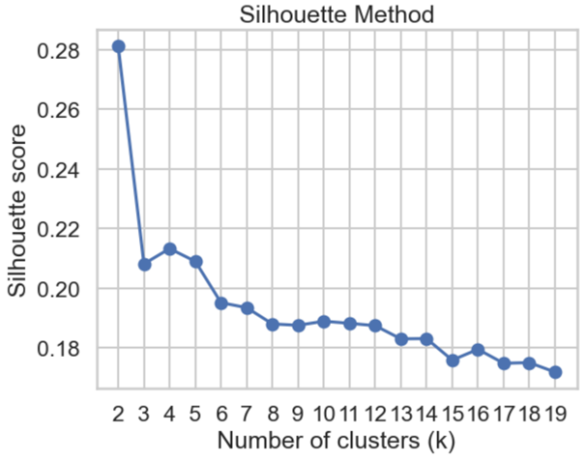

Figure 3 - Silhouette Method

The silhouette score was highest at k = 2, with a comparable value at k = 3, after which scores declined steadily. Although k = 2 provides the strongest geometric separation, a three-cluster solution was selected to balance cluster quality with interpretability and marketing relevance.

Hierarchical Clustering

Hierarchical clustering using Ward linkage revealed a small number of dominant branches with clear separation at higher linkage distances, followed by moderate-distance secondary splits. This structure supports two broad customer groups with an additional subdivision, consistent with a three-cluster solution.

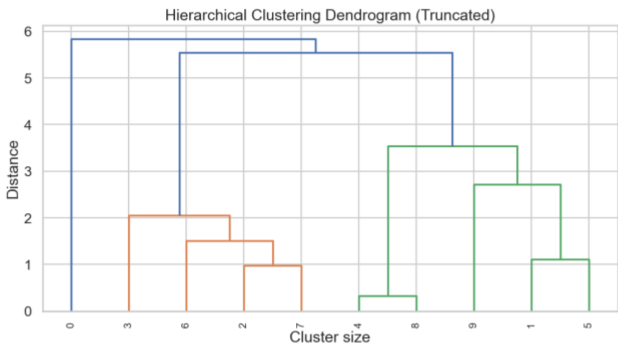

Figure 4 - Hierarchical Clustering Dendrogram

K-Means Clustering

K-means clustering was applied to the transformed and standardised feature set using Euclidean distance. The model was implemented using a scikit-learn pipeline with k = 3, random_state = 42, and n_init = "auto" to ensure reproducibility and robust centroid initialisation.

Results

Cluster Boxplots

The k-means clustering model with k = 3 assigned customers into three segments (Table 3).

Cluster 0 Customers	Cluster 1 Customers	Cluster 2 Customers
12294 (18%)	27723 (41%)	28283 (41%)

Table 3 - Cluster Apportionment

The model identified three customer segments with clear behavioural separation (Table 3). Boxplots show Cluster 0 as low engagement and value, Cluster 2 as high engagement and value, and Cluster 1 as an intermediate, heterogeneous group (Table 4).

Feature	Cluster 0	Cluster 1	Cluster 2
Frequency	Low	Medium	High
Recency	High	Medium	Low
CLV	Low	Medium	High
Average Unit Cost	Low	High	Medium
Customer Age	High	Medium	Low

Table 4 - Summary of Cluster Characteristics

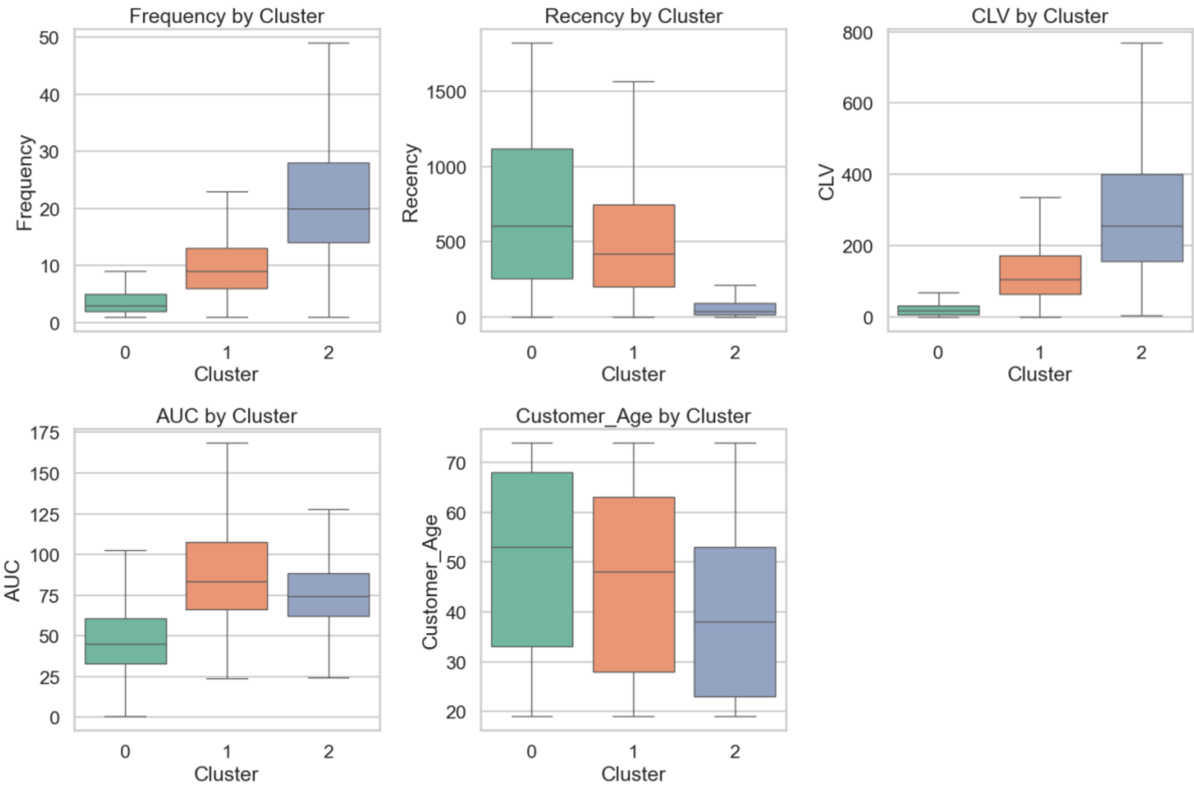

Figure 5 - Boxplot of the Unscaled Data Coloured by Cluster Number

Principal Component Analysis (PCA)

PCA reveals three broad customer groups with partial separation primarily along the first principal component, alongside overlap at cluster boundaries, indicating linear variance alone is insufficient in separating customer behaviour, motivating the use of nonlinear techniques.

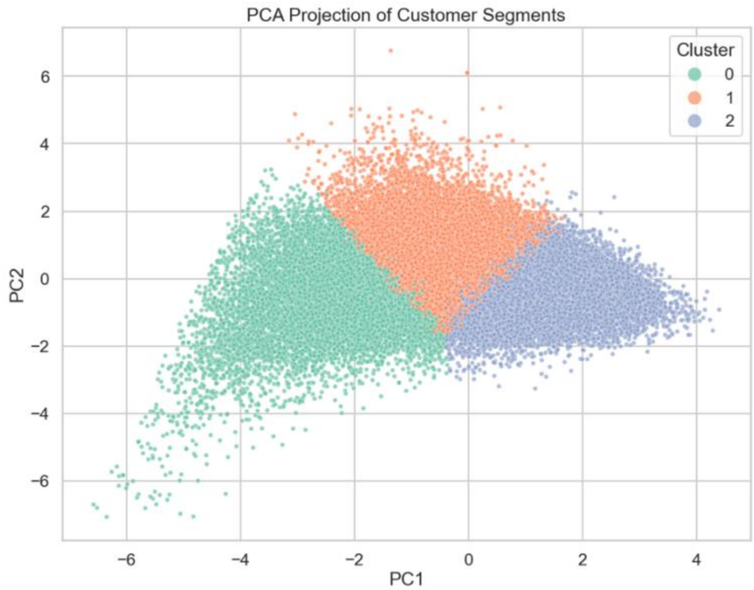

Figure 6 – PCA Projection of the K-Means Cluster Segments

t-Distributed Stochastic Neighbour Embedding (t-SNE)

t-SNE highlights pronounced local neighbourhood structure within each customer segment, revealing multiple dense subgroups separated by low-density regions. Transitions between segments are gradual rather than discrete, indicating continuous variation in customer behaviour. The improved local separation relative to PCA suggests that customer segments are differentiated by nonlinear relationships.

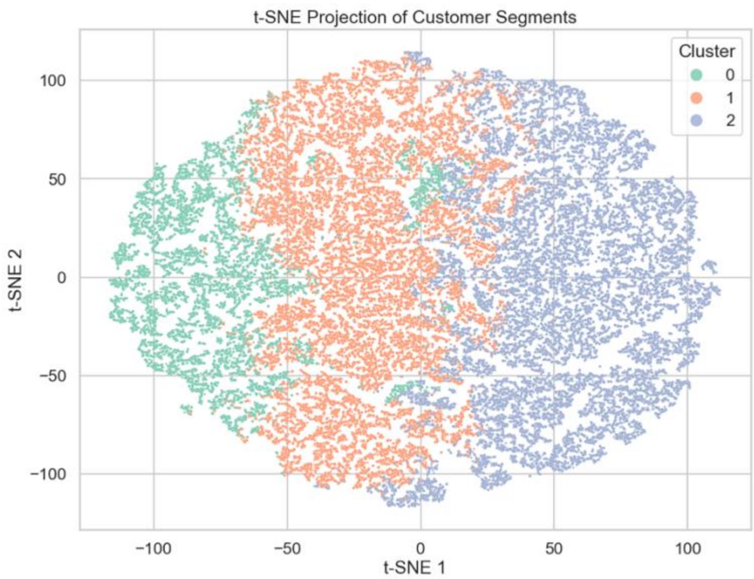

Figure 7 – T-SNE Projection of the K-Means Cluster Segments

Uniform Manifold Approximation and Projection (UMAP)

The UMAP projection reinforces the t-SNE findings while additionally revealing clear large-scale separation between customer segments, supporting the robustness of the three-cluster solution. The presence of internal sub-grouping indicates meaningful behavioural heterogeneity within clusters. Compared with t-SNE, UMAP provides stronger insight into the global organisation of customer behaviour while preserving local neighbourhood structure, improving the interpretability and stability of the segmentation.

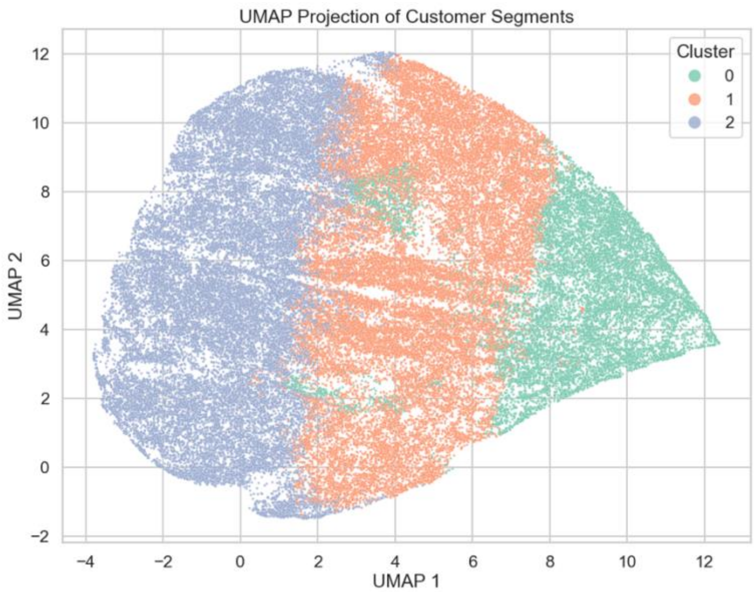

Figure 8 - UMAP Projection of the K-Means Cluster Segments

Next Steps

- Assess cluster stability under alternative initialisations and subsampling
- Evaluate the temporal stability of segments across time periods
- Formulate testable, domain-validated marketing hypotheses based on identified segments